

Koiter asymptotic analysis about a non-linear pre-buckling state

Implementation notes for `NLprebuck=True` in `bfscylinder_models`

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Abstract

This note documents the implementation of the single-mode Koiter method with a non-linear pre-buckling state in `koiter_cylinder_newton_raphson.py` (Donnell-Mushtari-Vlasov kinematics) and `koiter_cylinder_newton_raphson_sanders.py` (Sanders-Koiter kinematics). It covers the tangent stiffness matrix and the consistency it must satisfy, the reduction of the pre-buckling problem to the axisymmetric subspace, the incremental-iterative scheme that places the expansion point in the neighbourhood of the bifurcation point, the removal of the arbitrariness left by the degenerate buckling modes, and the stiffness-weighted orthogonality condition imposed on the second-order fields. The notation follows the finite element chapter of *Stability and Vibrations of Thin-Walled Composite Structures*, 2nd edition. Equation numbers of the form “Sun Eq. (nn)” refer to Sun, Tian, Li and Wang, *Accelerated Koiter method for post-buckling analysis of thin-walled shells under axial compression*, Thin-Walled Structures **155** (2020) 106962.

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1 Scope and notation

The routine `fkoiter_cyl_SS3` performs, for a circular cylindrical shell under axial compression with SS-3 boundary conditions,

1. a pre-buckling analysis,
2. a buckling eigenvalue analysis about the pre-buckling state,
3. the Koiter asymptotic expansion about the bifurcation point, returning the post-buckling coefficients a_{ijk} and b_{ijkl} .

With `NLprebuck=False` the pre-buckling state is the linear elastic solution and the pre-buckling path is exactly linear in the load parameter. With `NLprebuck=True` the pre-buckling state is obtained from a geometrically non-linear analysis, and the whole expansion must then be carried out about a state *in the neighbourhood of the bifurcation point*. This document describes the latter case.

The shell coordinates are the axial x and the circumferential θ , the radius is r and the thickness h . The degrees-of-freedom are collected in $\bar{\mathbf{u}}$, the fundamental (pre-buckling) state is $\hat{\mathbf{u}}$, and the buckling modes are $\delta\bar{\mathbf{u}}_i$. The external load is proportional, $\mathbf{F}_{ext} = \lambda\mathbf{F}_{ref}$, with $\mathbf{F}_{ref}$ the reference load vector built from N_{xx}^{unit} . Two load levels are distinguished: λ_b , at which the pre-buckling state is evaluated, and λ_c , the bifurcation load.

2 Kinematics

The membrane strains are written, for both kinematic models, as

$$\boldsymbol{\varepsilon} = \mathbf{B}_m \bar{\mathbf{u}} + \frac{1}{2} \begin{Bmatrix} g_x^2 \\ g_\theta^2 \\ 2g_x g_\theta \end{Bmatrix}, \quad \boldsymbol{\kappa} = \mathbf{B}_b \bar{\mathbf{u}} \quad (1)$$

where g_x and g_θ are the two gradient measures that carry the geometric non-linearity:

$$\text{Donnell-Mushtari-Vlasov: } g_x = w_{,x}, \quad g_\theta = w_{,\theta}, \quad \text{Sanders-Koiter: } g_x = w_{,x}, \quad g_\theta = w_{,\theta} - \frac{v}{r} \quad (2)$$

Every non-linear term of the Sanders-Koiter membrane strains is recovered from the Donnell-Mushtari-Vlasov one by the substitution $(w_{,x}, w_{,\theta}) \rightarrow (g_x, g_\theta)$, which is why the two modules differ only in the definition of Eq. 2 and in the corresponding interpolation matrices $\mathbf{G}_x$ and $\mathbf{G}_\theta$, with $g_x = \mathbf{G}_x \bar{\mathbf{u}}$ and $g_\theta = \mathbf{G}_\theta \bar{\mathbf{u}}$. The additional term $-v/r$ of the Sanders-Koiter operator $\mathbf{G}_\theta$ is the same one highlighted in the geometric stiffness matrix of the linear buckling chapter. The constitutive relation is the usual laminated-shell one, collectively denoted $\mathbf{C}$, relating $\{\mathbf{N}, \mathbf{M}\}$ to $\{\boldsymbol{\varepsilon}, \boldsymbol{\kappa}\}$ through the ABD matrix.

2.1 State and rates are independent fields

The pre-buckling state is a curve $\hat{\mathbf{u}}(\lambda)$. The asymptotic expansion needs, besides the state itself, its first and second derivatives with respect to λ , written $\hat{\mathbf{u}}_{,\lambda}$ and $\hat{\mathbf{u}}_{,\lambda\lambda}$. Differentiating Eq. 1 along the path, and abbreviating $\hat{g}_\alpha = \mathbf{G}_\alpha \hat{\mathbf{u}}$, $g_{\alpha,\lambda} = \mathbf{G}_\alpha \hat{\mathbf{u}}_{,\lambda}$, $g_{\alpha,\lambda\lambda} = \mathbf{G}_\alpha \hat{\mathbf{u}}_{,\lambda\lambda}$, with $\alpha \in \{x, \theta\}$:

$$\boldsymbol{\varepsilon}_{,\lambda} = \mathbf{B}_m \hat{\mathbf{u}}_{,\lambda} + \begin{Bmatrix} \hat{g}_x g_{x,\lambda} \\ \hat{g}_\theta g_{\theta,\lambda} \\ \hat{g}_x g_{\theta,\lambda} + \hat{g}_\theta g_{x,\lambda} \end{Bmatrix}, \quad \boldsymbol{\kappa}_{,\lambda} = \mathbf{B}_b \hat{\mathbf{u}}_{,\lambda} \quad (3)$$

$$\boldsymbol{\varepsilon}_{,\lambda\lambda} = \mathbf{B}_m \hat{\mathbf{u}}_{,\lambda\lambda} + \begin{Bmatrix} g_{x,\lambda}^2 + \hat{g}_x g_{x,\lambda\lambda} \\ g_{\theta,\lambda}^2 + \hat{g}_\theta g_{\theta,\lambda\lambda} \\ 2g_{x,\lambda} g_{\theta,\lambda} + \hat{g}_x g_{\theta,\lambda\lambda} + \hat{g}_\theta g_{x,\lambda\lambda} \end{Bmatrix}, \quad \boldsymbol{\kappa}_{,\lambda\lambda} = \mathbf{B}_b \hat{\mathbf{u}}_{,\lambda\lambda} \quad (4)$$

and the derivatives with respect to the degrees-of-freedom:

$$\frac{\partial \boldsymbol{\varepsilon}}{\partial \bar{\mathbf{u}}} = \mathbf{B}_m + \begin{Bmatrix} \hat{g}_x \mathbf{G}_x \\ \hat{g}_\theta \mathbf{G}_\theta \\ \hat{g}_x \mathbf{G}_\theta + \hat{g}_\theta \mathbf{G}_x \end{Bmatrix}, \quad \frac{\partial^2 \boldsymbol{\varepsilon}}{\partial \lambda \partial \bar{\mathbf{u}}} = \begin{Bmatrix} g_{x,\lambda} \mathbf{G}_x \\ g_{\theta,\lambda} \mathbf{G}_\theta \\ g_{x,\lambda} \mathbf{G}_\theta + g_{\theta,\lambda} \mathbf{G}_x \end{Bmatrix} \quad (5)$$

If one assumes the pre-buckling path to be linear in the load parameter, $\hat{\mathbf{u}}(\lambda) = \lambda \hat{\mathbf{u}}(1)$, then $\hat{\mathbf{u}}_{,\lambda} = \hat{\mathbf{u}}(1)$, $\hat{\mathbf{u}}_{,\lambda\lambda} = \mathbf{0}$ and $\hat{g}_\alpha = \lambda g_{\alpha,\lambda}$, so Eqs. 3 to 5 collapse onto expressions containing λ explicitly. That assumption is precisely what removes the non-linear pre-buckling behaviour from the expansion, and it is not made here: state, first rate and second rate are carried as three independent nodal fields.

3 The tangent stiffness matrix

3.1 Definition and decomposition

The stiffness matrix used throughout is the derivative of the internal force vector with respect to the degrees-of-freedom. Differentiating the residual $\mathbf{R} = \mathbf{F}_{int} - \mathbf{F}_{ext}$, and with no follower forces, so that $\partial \mathbf{F}_{ext} / \partial \bar{\mathbf{u}} = \mathbf{0}$:

$$\mathbf{R}' = \frac{\partial \mathbf{R}}{\partial \bar{\mathbf{u}}} = \frac{\partial \mathbf{F}_{int}}{\partial \bar{\mathbf{u}}} = \mathbf{K}_T \quad (6)$$

Differentiating the internal force vector produces one contribution from the change of the stresses and another from the change of the differentiation operator itself:

$$\mathbf{K}_T = \underbrace{\int_{\Omega} (\mathbf{B}_L + \mathbf{B}_{NL})^\top \mathbf{C} (\mathbf{B}_L + \mathbf{B}_{NL}) d\Omega}_{\mathbf{K}_C} + \underbrace{\int_{\Omega} \frac{\partial (\mathbf{B}_L + \mathbf{B}_{NL})^\top}{\partial \bar{\mathbf{u}}} \boldsymbol{\sigma} d\Omega}_{\mathbf{K}_G} = \mathbf{K}_C + \mathbf{K}_G \quad (7)$$

For the elements used here the constitutive contribution is split further into the constant part and the part that depends on the displacement state:

$$\mathbf{K}_C(\bar{\mathbf{u}}) = \mathbf{K}_0 + \mathbf{K}_C^{NL}(\bar{\mathbf{u}}), \quad \mathbf{K}_0 = \int_{\Omega} \mathbf{B}_L^\top \mathbf{C} \mathbf{B}_L d\Omega \quad (8)$$

which matches the three assembly routines of the element library, `update_KC0`, `update_KCNL` and `update_KG`, so that $\mathbf{K}_T = \mathbf{K}_0 + \mathbf{K}_C^{NL} + \mathbf{K}_G$. The internal force vector is assembled separately, by `update_fint`.

3.2 Verifying consistency

Nothing in the assembly guarantees that the three matrices add up to the derivative of the separately assembled $\mathbf{F}_{int}$; that has to be verified. The test is a directional Taylor expansion. For an arbitrary direction $\mathbf{d}$ and a step ϵ :

$$e(\epsilon) = \frac{\|\mathbf{F}_{int}(\bar{\mathbf{u}} + \epsilon\mathbf{d}) - \mathbf{F}_{int}(\bar{\mathbf{u}}) - \epsilon\mathbf{K}_T\mathbf{d}\|}{\|\epsilon\mathbf{K}_T\mathbf{d}\|} \quad (9)$$

The numerator is $O(\epsilon^2)$ when $\mathbf{K}_T$ is the derivative, so $e(\epsilon)$ must fall proportionally to ϵ , dividing by ten at every decade, until round-off takes over. A plateau means the matrix is not the derivative. Measured on a single element with a displacement state of the order of the thickness:

ϵ	before <code>bfscylinder</code> 0.6.0	from 0.6.0 onwards
10^{-2}	6.186×10^{-2}	9.466×10^{-4}
10^{-3}	6.157×10^{-2}	9.470×10^{-5}
10^{-4}	6.154×10^{-2}	9.470×10^{-6}
10^{-5}	6.154×10^{-2}	9.470×10^{-7}
10^{-6}	6.154×10^{-2}	9.450×10^{-8}

A second test identifies where a deficiency sits. Taking the true directional derivative $\mathbf{Jd}$ from a central difference and fitting the coefficients of the two state-dependent matrices by least squares,

$$\min_{c_1, c_2} \|\mathbf{K}_0\mathbf{d} + c_1\mathbf{K}_C^{NL}\mathbf{d} + c_2\mathbf{K}_G\mathbf{d} - \mathbf{Jd}\| \quad (10)$$

returns $c_1 = c_2 = 1.0000$ with a residual of 7.6×10^{-11} from 0.6.0 onwards. Before 0.6.0 it returned $c_1 = 1.5122$ and $c_2 = 1.2876$ with a residual of 3.8×10^{-2} , against 6.0×10^{-2} for the assembled combination: no recombination of the three matrices reproduced the derivative, which identified the defect as a missing term rather than a mis-scaled one. Both kinematics are covered, the Sanders-Koiter element reaching 1.2×10^{-10} .

3.3 Why the consistency matters twice

An inconsistent tangent stiffness matrix has two distinct consequences here, and only the first is the familiar one.

In the Newton-Raphson solution of Section 4 the tangent controls the rate of convergence only, the converged state being fixed exclusively by the residual, that is by the accuracy with which $\mathbf{F}_{int}$ and $\mathbf{F}_{ext}$ are computed. An inconsistent tangent therefore degrades the iteration from quadratic to linear, with a contraction factor that grows as the tangent is approached, but it does not by itself move the equilibrium state. On the coarse meshes of Section 11 the contraction was about 0.3 per iteration, which still converges; on a mesh with $n_\theta = 60$ it reached 0.95, exhausting the iteration budget at every load step and leaving the load stepping stalled near $\lambda_b/\lambda_c = 0.94$.

The second consequence has no such protection. The matrix $\mathbf{K}_C = \mathbf{K}_0 + \mathbf{K}_C^{NL}$ does not only drive the iteration: it enters the eigenvalue problem of Eq. 12 and the expansion operator of Eq. 20 directly, where it is not a preconditioner but part of the statement of the problem. An error in $\mathbf{K}_C^{NL}$ is therefore inherited by λ_c , by the buckling mode and by every post-buckling coefficient. Correcting it moved the buckling loads of Section 11 by 9 to 11 per cent, onto the reference solutions, and reversed the sign of b_{1111} .

4 The axisymmetric pre-buckling solver

4.1 The fundamental path is axisymmetric

The fundamental path of a perfect cylinder under uniform axial compression is axisymmetric, and every buckling mode that makes $\mathbf{K}_T$ singular is not. Solving the pre-buckling problem in the axisymmetric subspace therefore keeps the iteration clear of the entire near-critical cluster, and not merely of the few modes that happen to have been computed. This is also what the reference solutions do, AXBIF and ANILISA both solving an axisymmetric pre-buckling problem.

4.2 Cyclic symmetry of the mesh

The mesh has n_θ elements around the circumference, and the element degrees-of-freedom are expressed in the local shell frame, which turns with θ . Shifting the circumferential node index cyclically by one element is therefore a symmetry of every operator assembled on the mesh: if $\mathbf{P}$ denotes that shift, then $\mathbf{P}^\top \mathbf{K} \mathbf{P} = \mathbf{K}$ for $\mathbf{K} \in \{\mathbf{K}_0, \mathbf{K}_C^{NL}, \mathbf{K}_G\}$ whenever the state they are evaluated at is itself invariant.

Two consequences are used. The first is that the average over the orbit of $\mathbf{P}$ is the orthogonal projector onto the invariant subspace, used to remove the round-off asymmetry of the starting state. The second, used in Section 6, is that $\mathbf{P}$ maps a buckling mode onto another buckling mode carrying the same multiplier.

4.3 The reduced problem

Let $\mathbf{B}_{axi}$ collect one column per axial station and per degree-of-freedom, carrying one at that degree-of-freedom of every node of the station, so that an axisymmetric field is $\mathbf{B}_{axi} \mathbf{a}$ and the columns are orthogonal with squared norm n_θ . The Newton correction is obtained from the Galerkin projection of the tangent problem onto that basis:

$$\left(\mathbf{B}_{axi}^\top \mathbf{K}_T \mathbf{B}_{axi} \right) \mathbf{a} = - \mathbf{B}_{axi}^\top \mathbf{R}, \quad \delta \bar{\mathbf{u}} = \mathbf{B}_{axi} \mathbf{a} \quad (11)$$

The reduced system has $n_x \times n_{DOF}$ unknowns against the full count, 250 against 15000 on the $n_\theta = 60$ mesh, and it removes the need to deflate the buckling modes from the correction, the whole near-critical cluster lying outside the subspace by construction. Because the residual of an axisymmetric state is itself axisymmetric, $\mathbf{B}_{axi}^\top \mathbf{R} = \mathbf{0}$ implies $\mathbf{R} = \mathbf{0}$ on the free degrees-of-freedom, so the reduced solution is an exact solution of the full equilibrium equations and not an approximation of one.

A reduced coordinate is treated as known as soon as one of the nodes it spans is constrained. This matters for the single-node constraint that suppresses the axial rigid body translation, which is the one part of the boundary conditions that is not rotationally symmetric: in the reduced problem it fixes the whole axial station, which is what an axisymmetric field requires of it anyway. The same asymmetry is handled in Section 6 by subtracting an axial rigid body translation, which carries no strain and is annihilated by both stiffness matrices.

The reduced matrix mixes nodal displacements with nodal derivatives, whose entries differ by orders of magnitude, and reaches a condition number of 4×10^{12} . The sparse solver used elsewhere equilibrates internally, the dense one does not, so the reduced matrix is scaled symmetrically before the solve, which brings the condition number to 3×10^2 . Measured on the $n_\theta = 60$ case this leaves every iterate unchanged to ten digits, the graded matrix being solved accurately by the pivoting alone; it is kept as a guard rather than as a correction.

5 Locating the expansion point

5.1 Why the load level matters

Koiter's expansion is asymptotic about the bifurcation point. Sun §2.2 requires the pre-buckling state to satisfy $\lambda_b/\lambda_c \approx 0.995$. If instead the pre-buckling problem is solved once at the reference load and the eigenvalue problem is posed about that state, then for the shells considered here λ_b/λ_c is of the order of 0.1. At that load level the radial pre-buckling deflection is one to three per cent of a wall thickness, the boundary layer produced by the edge restraint has not developed, and $\|\mathbf{K}_C^{NL}(\hat{\mathbf{u}})\|/\|\mathbf{K}_0\| \sim 2 \times 10^{-4}$; the eigenvalue analysis is then a linear buckling analysis in all but name and the whole computation returns the *membrane* pre-buckling answer.

5.2 Iterative eigenvalue algorithm

The expansion point is located by the incremental-iterative scheme of Sun Eqs. (44) to (46). At the current load level λ_b the pre-buckling state $\hat{\mathbf{u}}$ is known and the eigenvalue problem

$$[\mathbf{K}_C(\hat{\mathbf{u}}) + \mu \mathbf{K}_G(\hat{\mathbf{u}})] \delta \bar{\mathbf{u}} = \mathbf{0} \quad (12)$$

gives the multiplier μ of the current stress state, so that $\lambda_c = \lambda_b \mu$. This is the eigenvalue problem of the linear buckling chapter with $\mathbf{K}_0$ replaced by the constitutive part of the tangent at the current state.

5.3 Advancing the load level

Sun Eq. (44) advances the load level by a fixed fraction of the distance to the bifurcation load:

$$\lambda_b \leftarrow \lambda_b + \eta (\lambda_c - \lambda_b) \quad (13)$$

with $\eta = 0.8$ on the first step and $\eta = 0.5$ afterwards. The difficulty is that λ_c is not fixed: it is re-evaluated at the new state and it falls as λ_b rises. Writing $s = d\lambda_c/d\lambda_b$ and linearising, the new distance to the bifurcation point is

$$\lambda'_b - \lambda'_c = (\lambda_c - \lambda_b) [\eta(1 - s) - 1] \quad (14)$$

so the step lands *past* the bifurcation point whenever

$$\eta > \frac{1}{1 - s} \quad (15)$$

The sensitivity s steepens as the bifurcation point is approached and as the mesh is refined. On the $n_\theta = 40$ meshes of Section 11 it reaches -0.25 and -0.72 , for which Eq. 15 permits $\eta = 0.80$ and $\eta = 0.58$, both above the fixed 0.5, and no overshoot occurs. On the $n_\theta = 60$ mesh it reaches -1.09 , permitting only $\eta = 0.478$, and the fixed $\eta = 0.5$ overshoots to $\lambda_b/\lambda_c = 1.0013$.

The implementation therefore estimates s from the two previous load steps and caps the step at a fraction f of the permissible value:

$$\eta \leftarrow \min \left(\eta, \frac{f}{1 - s} \right), \quad s = \frac{\lambda_c - \lambda_c^{prev}}{\lambda_b - \lambda_b^{prev}}, \quad f = 0.7 \quad (16)$$

Substituting Eq. 16 into Eq. 14 gives $\lambda'_b - \lambda'_c = -(1 - f)(\lambda_c - \lambda_b)$ whatever the value of s , so the cap does not merely prevent the overshoot: it turns Eq. 13 into a secant iteration on $\lambda_c(\lambda_b) - \lambda_b$, which divides the remaining distance by a fixed factor at every step.

The fraction is 0.7 rather than something closer to unity because s is estimated backwards and lags while it steepens. On the $n_\theta = 60$ mesh the offending step is taken with $s = -0.66$ measured over the previous interval, which permits $\eta = 0.54$ and therefore does not bind on the fixed $\eta = 0.5$, while the step itself turns out to have $s = -1.09$. A margin of about a quarter on $1 - s$ absorbs that lag, and with $f = 0.7$ the same mesh lands at $\lambda_b/\lambda_c = 0.9965$ instead of 1.0013, in the same number of load steps.

5.4 Accepting the state

Convergence is declared by Sun Eq. (45). That criterion measures the distance to the bifurcation point from below only, so that once λ_b passes λ_c its left-hand side turns negative and it accepts the state whatever the overshoot. The distance is therefore taken in absolute value:

$$\frac{|\lambda_c - \lambda_b|}{\lambda_c} \leq \varepsilon_1, \quad \varepsilon_1 = 0.005 \quad (17)$$

If the load level lands past the bifurcation point by more than ε_1 , the load step is taken again from the previous state with η halved. The fundamental path is still computable there, the pre-buckling solve being restricted to the axisymmetric subspace, but the expansion is meant to be made on the near side of it.

At convergence $\mu \simeq 1$ to within ε_1 , which is what makes the operators of Eq. 12 usable directly in the expansion.

5.5 Rates of the pre-buckling state

The first rate follows from Sun Eq. (29), using the tangent stiffness at the converged pre-buckling state and solved in the reduced basis of Eq. 11:

$$\mathbf{K}_T \hat{\mathbf{u}}_{,\lambda} = \mathbf{F}_{ref} \quad (18)$$

The second rate follows from Sun Eq. (35) as a backward difference against the load step preceding the converged one:

$$\hat{\mathbf{u}}_{,\lambda\lambda} = \frac{\hat{\mathbf{u}}_{,\lambda} - \hat{\mathbf{u}}_{,\lambda}^{prev}}{\lambda_b - \lambda_b^{prev}} \quad (19)$$

Reusing a step the load stepping has already converged on costs nothing and, unlike a step solved ahead of λ_b , cannot fall on the far side of the bifurcation point. The corresponding stress rates, Sun Eqs. (30) and (34), follow from Eqs. 3 and 4 through the constitutive matrix.

6 Buckling modes of a cylinder

6.1 The modes come in degenerate pairs

For a circular cylinder the buckling modes occur in degenerate pairs, one for each sign of the circumferential wave number, so $\dim \ker \geq 2$ even when a single Koiter mode is retained, and every rotation of a pair inside its own eigenspace is again a pair of buckling modes. Which member of it the eigen solver returns is decided by round-off, and changes with the BLAS implementation.

That would not matter if the post-buckling coefficients were invariant under that rotation, and they are not. The second-order field of a mode with n circumferential waves carries the harmonic $2n$, which a mesh sized for the buckling mode itself barely resolves. Measured on the case of Section 11 with $n_\theta = 40$ and $n = 10$, rotating the critical pair by 30 degrees moves b_{1111} from -0.230556 to -0.192752 , sixteen per cent, with the pre-buckling state and the buckling load unchanged to eleven digits. Before the tangent of Section 3 was made consistent the same rotation moved it from -0.0008 to $+0.2744$, sign included, so most of that dependence was the tangent and what remains is the mesh.

6.2 Fixing the member

Each pair is rotated to the member whose crest falls on the $\theta = 0$ generator, which is a property of the mesh and not of the arithmetic. Of the two members of a pair one has a crest on that

generator and the other a node, so the dominant eigenvector of the 2×2 Gram matrix of their traces there is well separated, and the sign is fixed by requiring the crest to point outwards.

The partner of a mode need not be among the ones the eigen solver returned. A Krylov method builds its subspace from a single starting vector, so it finds one direction of a degenerate eigenspace and then moves on to the next multiplier: asking for the two lowest multipliers of the shell of Section 11 returns one member of the critical pair and one member of the next pair, 3.3×10^{-4} away. A missing partner is recovered through the cyclic shift $\mathbf{P}$ of Section 4, which maps a mode onto another mode of the same pair, after removal of the axial rigid body translation that restores the single-node constraint. Truncating the offending degree-of-freedom instead takes the mode out of its eigenspace and, the operator of the second-order field being singular along it, the error is then amplified without bound.

This makes the coefficients reproducible: 9×10^{-9} between a single threaded and a multi threaded run, against a sign difference between operating systems before. It does not make them mesh converged, and only a mesh that resolves the harmonic $2n$ would.

7 Consistency of the operators

The operator appearing in the asymptotic expansion,

$$\phi_2 = \mathbf{K}_C(\hat{\mathbf{u}}) + \mu \mathbf{K}_G(\hat{\mathbf{u}}) \quad (20)$$

must be *the same* operator whose null vector is the buckling mode, that is, exactly the operator of the eigenvalue problem of Eq. 12. Building ϕ_2 from a differently evaluated $\mathbf{K}_C^{NL}$ than the one used there, for instance from $\mathbf{K}_C^{NL}(\lambda_c \hat{\mathbf{u}})$ while the eigenvalue problem used $\mathbf{K}_C^{NL}(\hat{\mathbf{u}})$, leaves ϕ_2 non-singular along $\delta \bar{\mathbf{u}}$ and corrupts every quantity derived from it. The severity is easy to measure: with the inconsistent pairing the expansion operator does not annihilate the buckling mode at all, the relative residual rising from 10^{-10} to 4.6. Since $\lambda_b \simeq \lambda_c$ after the iterative algorithm has converged, $\mu \simeq 1$ and Eq. 20 is simply the tangent operator at the critical state.

The discrete energy derivatives assembled element by element are the building blocks of the expansion:

symbol in the code	meaning
phi2	$\phi_2 = \partial^2 \phi / \partial \bar{\mathbf{u}}^2$, the tangent operator at the critical state
phi3_ab[(i,j)]	ϕ_3 contracted with modes i, j , one free index
phi4[(i,j,k,l)]	ϕ_4 fully contracted
phi20_a[i]	$\partial \phi_2 / \partial \lambda$ contracted with mode i , one free index
phi30_ab[(i,j)]	$\partial \phi_3 / \partial \lambda$ contracted with modes i, j , one free index
phi200_ab[(i,j)]	$\partial^2 \phi_2 / \partial \lambda^2$ fully contracted

All of them are built from Eqs. 3 to 5 together with the constant quadratic form $\partial^2 \varepsilon / \partial \bar{\mathbf{u}} \partial \bar{\mathbf{u}}$, which for Eq. 1 is $\mathbf{G}_x^\top \mathbf{G}_x$, $\mathbf{G}_\theta^\top \mathbf{G}_\theta$ and $\mathbf{G}_x^\top \mathbf{G}_\theta + \mathbf{G}_\theta^\top \mathbf{G}_x$ in the three strain components.

8 Second-order fields and the orthogonality condition

8.1 The singular system

The second-order fields $\bar{\mathbf{u}}_{ij}$ solve

$$\phi_2 \bar{\mathbf{u}}_{ij} = \mathbf{g}_{ij}, \quad \mathbf{g}_{ij} = -\frac{1}{2} \phi_{3,ij} - \frac{1}{m} \sum_k a_{kij} \lambda_k \phi_{20,k} \quad (21)$$

with m the number of Koiter modes. By construction ϕ_2 is singular, its null space being spanned by the buckling modes. Consequently Eq. 21 determines $\bar{\mathbf{u}}_{ij}$ only up to an arbitrary element of that null space, and a plain sparse solve followed by a projection is not a well-posed procedure. Two things are therefore needed: a solver that copes with the singularity, and the correct condition that fixes the null-space component.

8.2 The condition to be imposed

The orthogonality condition of Koiter's theory is Sun Eq. (18), whose finite element form is Sun Eq. (33):

$$\delta \bar{\mathbf{u}}_k^\top [\mathbf{K}_D(\hat{\mathbf{u}}, \hat{\mathbf{u}}_{,\lambda}) + \mathbf{K}_G(\boldsymbol{\sigma}_{,\lambda})] \bar{\mathbf{u}}_{ij} + \frac{1}{2} \delta \bar{\mathbf{u}}_k^\top \mathbf{B}_{NL}^\top (\delta \bar{\mathbf{u}}_k) \mathbf{C} \mathbf{B}_{NL} (\delta \bar{\mathbf{u}}_k) \hat{\mathbf{u}}_{,\lambda} = 0 \quad (22)$$

Two observations turn this into something directly computable. The bracket in the first term is the load-parameter derivative of the tangent operator, $\partial \phi_2 / \partial \lambda$, so the first term is exactly the contraction stored as `phi20_a[k]`. The second term is a constant, which follows from ϕ_{30} without any new element integral: $\phi_{30,ij}$ is assembled as one half of the quadrature weight times the sum of six contractions which, when all mode indices coincide, are equal to one another by the symmetry of the constitutive matrix and of the quadratic form. Hence

$$c_k = \frac{\phi_{30,ij} \cdot \delta \bar{\mathbf{u}}_k}{6} \quad (23)$$

and the condition to impose is

$$\phi_{20,k} \cdot \bar{\mathbf{u}}_{ij} = -c_k \quad (24)$$

It is weighted by the stiffness matrices, through ϕ_{20} , and inhomogeneous, through c_k . A Gram-Schmidt projection, or equivalently a symmetric Euclidean border, imposes neither of these properties and is the procedure that Eq. 24 replaces.

8.3 The bordered system

Both requirements are met by a *non-symmetric* bordered system:

$$\begin{bmatrix} \phi_2 & \mathbf{N}_{sp} \\ \mathbf{W}^\top & \mathbf{0} \end{bmatrix} \begin{Bmatrix} \bar{\mathbf{u}}_{ij} \\ \boldsymbol{\alpha} \end{Bmatrix} = \begin{Bmatrix} \mathbf{g}_{ij} \\ -\mathbf{c} \end{Bmatrix} \quad (25)$$

in which the roles of the two borders are distinct. The *column* border $\mathbf{N}_{sp}$ spans the null space of ϕ_2 and is what makes the system non-singular; the *row* border $\mathbf{W}$ carries the constraints, one column per Koiter mode, $\mathbf{W}[:,k] = \phi_{20,k}$, with the corresponding right-hand side $-c_k$ from Eq. 23. The system is non-singular provided $\mathbf{W}^\top \mathbf{N}_{sp}$ is, and its leading entry $\phi_{20,k} \cdot \delta \bar{\mathbf{u}}_k$ is exactly the quantity appearing in the denominator of the post-buckling coefficients, hence non-zero by construction.

Because the critical eigenspace is degenerate, $\mathbf{N}_{sp}$ must span the *whole* near-null space and not only the retained Koiter modes: every eigenvector whose multiplier equals the critical one within a relative tolerance is included, and the set is orthonormalised with a rank-revealing factorisation. Equation 22 supplies one condition per Koiter mode, which is fewer than the dimension of the null space; the degenerate partners lie outside single-mode theory, and the remaining rows of $\mathbf{W}$ keep the Euclidean condition. This is a choice, not a result; the consistent treatment of a degenerate critical mode is the multi-mode Koiter expansion.

8.4 What the weighted condition does, and does not, change

Condition 24 is the correct one and replaces a Gram-Schmidt projection that was neither weighted nor inhomogeneous. For the cases considered here, however, it changes almost nothing. Measured before the tangent was made consistent, running the same analysis under the two conditions with everything else identical moved the second-order field by 6.5×10^{-7} relative, so the two conditions are numerically near-coincident for this problem: the null-space component of $\bar{\mathbf{u}}_{ij}$ on which they disagree is small to begin with.

A second reason reinforces this. Condition 24 fixes only the component of $\bar{\mathbf{u}}_{ij}$ along $\ker \phi_2$, and b_{ijkl} sees $\bar{\mathbf{u}}_{ij}$ through the contractions $\phi_{3,ij} \cdot \bar{\mathbf{u}}_{kl}$; but $\phi_{3,ij}$ contracted with a buckling mode is precisely the numerator of a_{ijk} in Eq. 26. For the symmetric bifurcation of a cylinder under axial compression a_{ijk} vanishes, measured values being of order 10^{-5} against a b_{ijkl} of order unity, so $\phi_{3,ij}$ is nearly orthogonal to the null space. For an asymmetric bifurcation, with $a_{ijk} \neq 0$, the condition would carry more weight.

9 Post-buckling coefficients

With the second-order fields available, the coefficients are

$$a_{ijk} = -\frac{1}{2\lambda_i} \frac{\phi_{3,ij} \cdot \delta \bar{\mathbf{u}}_k}{\phi_{20,i} \cdot \delta \bar{\mathbf{u}}_i} \quad (26)$$

$$b_{ijkl} = -\frac{1}{6\lambda_i(\phi_{20,i} \cdot \delta \bar{\mathbf{u}}_i)} \left[\phi_{4,ijkl} + 3\phi_{3,ij} \cdot \bar{\mathbf{u}}_{kl} + 3\phi_{3,il} \cdot \bar{\mathbf{u}}_{jk} \right. \\ \left. + \lambda_i(a_{iij}\phi_{30,ik} \cdot \delta \bar{\mathbf{u}}_l + a_{ijk}\phi_{30,il} \cdot \delta \bar{\mathbf{u}}_i + a_{ikl}\phi_{30,ii} \cdot \delta \bar{\mathbf{u}}_j) \right. \\ \left. + \phi_{200,ii}\lambda_i^2(a_{iij}a_{ikl} + a_{ijk}a_{ili} + a_{ikl}a_{iij}) \right] \quad (27)$$

The buckling modes are normalised so that the largest translational amplitude equals the shell thickness, which is the normalisation used by the reference solutions.

10 Algorithm

Algorithm 1 Koiter analysis about a non-linear pre-buckling state

```

1: solve  $\mathbf{K}_0 \bar{\mathbf{u}} = \mathbf{F}_{ref}$ ; project onto the axisymmetric subspace;  $\lambda_b \leftarrow 1$ 
2: for  $i = 1, \dots, i_{\max}$  do
3:   assemble  $\mathbf{K}_C(\hat{\mathbf{u}})$ ,  $\mathbf{K}_G(\hat{\mathbf{u}})$ 
4:   solve Eq. 12  $\rightarrow \mu, \delta \bar{\mathbf{u}}$ ; canonicalise the degenerate pairs (Section 6);  $\lambda_c \leftarrow \lambda_b \mu$ 
5:   if Eq. 17 holds then break
6:   if  $\lambda_b > \lambda_c$  then restore the previous step, halve the cap on  $\eta$ , continue
7:    $\eta \leftarrow 0.8$  if  $i = 1$  else 0.5, capped by Eq. 16
8:   repeat
9:      $\lambda_b^{new} \leftarrow \lambda_b + \eta(\lambda_c - \lambda_b)$ 
10:    predictor  $\hat{\mathbf{u}} + \hat{\mathbf{u}}_{,\lambda}(\lambda_b^{new} - \lambda_b)$ ; Newton-Raphson in the reduced basis (Eq. 11)
11:    on failure  $\eta \leftarrow \eta/2$ 
12:  until converged or  $\eta$  exhausted
13:  store  $(\lambda_b, \lambda_c, \hat{\mathbf{u}}_{,\lambda})$  as the previous step; advance
14: end for
15:  $\hat{\mathbf{u}}_{,\lambda}$  from Eq. 18;  $\hat{\mathbf{u}}_{,\lambda\lambda}$  from Eq. 19
16: assemble  $\phi_2, \phi_3, \phi_4, \phi_{20}, \phi_{30}, \phi_{200}$ 
17:  $a_{ijk}$  from Eq. 26
18:  $\bar{\mathbf{u}}_{ij}$  from the bordered system Eq. 25 with the weighted condition Eq. 24
19:  $b_{ijkl}$  from Eq. 27

```

11 Effect on the results

The two verification cases are the composite cylinder of Sun §3.1 and the NASA shell AW-CYL-1-1 of Arbocz, Starnes and Nemeth (AIAA-2001-1392); they are the same shell, so the ANILISA n -search of the latter provides reference values for both. Buckling loads are normalised by $N_{cl} = E_{11}h^2 / (r\sqrt{3(1 - \nu_{12}^2)})$.

11.1 Buckling loads

	n	λ_c	reference
Arbocz, $n_\theta = 40$	10	0.331413	0.329163 (ANILISA, $n = 10$)
Arbocz, $n_\theta = 60$	11	0.330603	0.328594 (ANILISA, $n = 11$)
			0.327759 (STAGS-A, $n = 11$)
Sun, $n_\theta = 40$	10	0.346127	—
Sun, $n_\theta = 50$	10	0.347244	—

The two Arbocz meshes agree with the ANILISA entry for their own circumferential wave number to 0.7 and 0.6 per cent respectively, and the finer one with the Level-3 STAGS-A value to 0.9 per cent. The membrane pre-buckling regime, which is what the computation returns when the expansion point is left at the reference load, gives 0.365992 instead.

11.2 Post-buckling coefficients

	n	b_{1111}	reference
Arbocz, $n_\theta = 40$	10	-0.335975	—
Arbocz, $n_\theta = 60$	11	-0.356457	-0.37605 (ANILISA, $n = 11$) -0.3772 (Sun, Table 2)
Sun, $n_\theta = 40$	10	-0.230556	—
Sun, $n_\theta = 50$	10	-0.222844	—

The $n_\theta = 60$ value is within five per cent of the ANILISA coefficient for the same mode. A negative b means an imperfection sensitive shell, which is what this one is.

11.3 What the tangent correction changed

Before the tangent stiffness matrix was made consistent (Section 3.2) the same meshes returned $\lambda_c = 0.302887$ and $b_{1111} = +0.410664$ for the Arbocz case: the buckling load was nine per cent low and the post-buckling coefficient had the wrong sign. The sign was at that time attributed to a single-mode expansion about a degenerate critical mode being unable to determine it, which was incorrect. Two meshes that did not even select the same critical mode, $n = 10$ at $n_\theta = 40$ against $n = 7$ at $n_\theta = 50$ for the Sun case, now select $n = 10$ on both and agree to 0.3 per cent in the buckling load and 3.4 per cent in b_{1111} .

The Newton-Raphson behaviour changed with it. Each load step now takes a single iteration, with residuals falling 5.1×10^{-5} , 1.1×10^{-5} , 1.2×10^{-6} , 5.4×10^{-8} , 1.5×10^{-9} , and no step back-off anywhere. The $n_\theta = 60$ mesh, which previously exhausted the iteration budget at every load step, converges in four of them.

12 Known limitations

1. **Degenerate critical modes.** Condition 24 supplies one equation per retained mode while the null space is larger, and the rotation inside the degenerate pair still moves b_{1111} by sixteen per cent on the $n_\theta = 40$ mesh. The canonicalisation of Section 6 makes the choice reproducible rather than arbitrary; it does not make it converged. Multi-mode Koiter is the consistent treatment.
2. **Resolution of the second-order field.** The second-order field carries the harmonic $2n$, which the meshes used here sample with about two nodes per wave. This, and not the expansion itself, is what limits b_{ijkl} ; the buckling load is far less affected.
3. **Symmetrisation of ϕ_{30} for distinct modes.** The identity of Eq. 23 was established for coinciding mode indices, which covers the single-mode case exercised here; the general multi-mode symmetrisation remains to be worked out.
4. **Load control.** The scheme of Section 5.3 now reaches the requested neighbourhood of the bifurcation point on all meshes tried, but it remains load controlled. The extrapolated eigenvalue problem of Sun Eq. (27), $[\mathbf{K}_T + (\lambda_c - \lambda_b) \partial \mathbf{K}_T / \partial \lambda] \delta \bar{\mathbf{u}} = \mathbf{0}$, would remove the need to sit on top of the bifurcation point at all.
5. **Quadrature.** Convergence with respect to the number of Gauss points has been verified: $n_{int} = 4$ gives b to within 0.4 per cent of $n_{int} = 5$ and 6, while $n_{int} = 2$ under-integrates and produces a rank-deficient element.